

AN EXPONENT-ONE OBSTRUCTION AND A 95/102 LOCAL SIEVE FOR A LENHART-ATTRIBUTED FAMILY OF EULER BRICKS

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ABSTRACT. The perfect-cuboid problem asks whether an Euler brick can also have an integral space diagonal. The global problem remains open. We study the Lenhart-attributed family with edges

$$(u^2 - w^2)(v^2 - w^2), \quad 4uvw^2, \quad 2uw(v^2 - w^2), \quad u^2 + v^2 = 5w^2.$$

After an integral parametrization of the conic, its squared space diagonal is an explicit binary form $F(r, s)$ of degree 16. We prove that, for $s = 1$ and $r \equiv 3, 4, 7$, or $11 \pmod{169}$, one has $v_{13}(F(r, 1)) = 1$. Thus a fixed prime gives an exponent-one obstruction on four infinite arithmetic progressions. Independently, exact quadratic residue tests at 23, 31, and 67 exclude 95/102 of the projective Chinese-remainder cells for primitive parameter pairs. Complete residue lists and two independent integer-arithmetic verifiers accompany the result. The conclusions apply to this parameter family only and do not settle the existence of a perfect cuboid.

1. INTRODUCTION

A *perfect cuboid* is a triple of positive integers (a, b, c) for which

$$(1) \quad a^2 + b^2, \quad a^2 + c^2, \quad b^2 + c^2, \quad a^2 + b^2 + c^2$$

are all squares. If only the first three quantities are squares, the triple is an *Euler brick*. Perfect cuboids have been studied through Pythagorean parametrizations, rational points on algebraic curves, and exact searches; the existence question is still open [3, 1, 7].

Recent preprints provide a useful structural viewpoint. Peschmann reduces a master parametrization to a family of genus-three curves with elliptic quotients, develops descent obstructions, and verifies a bounded parameter region [5].

A companion computation reports 1,284,670 exact nonsquare tests, including 1,222,841 master tuples generated from Mordell–Weil groups. In the 151,575 records whose relevant norm was fully factored, every record

Date: August 22, 2026.

2020 Mathematics Subject Classification. Primary 11D09; Secondary 11Y50.

Key words and phrases. perfect cuboid, Euler brick, local obstruction, quadratic residue, exponent-one blocker.

has a prime divisor of exponent exactly one; the universal statement is explicitly left as a conjecture [4].

A separate torsion-intersection argument proves nonexistence on 1,072 explicit fibers with $\max(m, n) \leq 100$, not on the full parameter space [6].

The present note turns that observed mechanism into an elementary uniform theorem on one classical subfamily. The family is conventionally attributed to William Lenhart, although the historical primary source for the attribution is uncertain; we therefore use the term *Lenhart-attributed*, following the cautious convention in [4, 2].

Our two results are complementary. The first is 13-adic: four residue classes modulo 13^2 force the prime 13 to occur exactly once in the squared space diagonal. The second uses quadratic characters at three different primes and removes exactly 95/102 of the locally primitive parameter cells. Both statements follow from finite exact certificates. Neither extrapolates from a height search, and neither asserts global nonexistence.

2. THE PARAMETRIZED FAMILY

For $r, s \in \mathbb{Z}$, put

$$\begin{aligned} u &= r^2 - 4rs - s^2, \\ v &= -2(r^2 + rs - s^2), \\ w &= r^2 + s^2. \end{aligned} \tag{2}$$

Direct expansion gives

$$u^2 + v^2 = 5w^2. \tag{3}$$

Define the signed edge expressions

$$A = (u^2 - w^2)(v^2 - w^2), \quad B = 4uvw^2, \quad C = 2uw(v^2 - w^2). \tag{4}$$

Absolute values are understood when these are regarded as lengths.

Lemma 2.1 (Face identities and space norm). *For the parameters in (2),*

$$A^2 + B^2 = (u^4 - 5u^2w^2 - 4w^4)^2, \tag{5}$$

$$A^2 + C^2 = ((v^2 - w^2)(u^2 + w^2))^2, \tag{6}$$

$$B^2 + C^2 = (2uw(v^2 + w^2))^2. \tag{7}$$

Moreover, the squared space diagonal factors as

$$\begin{aligned} F(r, s) &:= A^2 + B^2 + C^2 = (u^2 + 4w^2) \\ &\quad \cdot (u^6 - 10u^4w^2 + 25u^2w^4 + 4w^6). \end{aligned} \tag{8}$$

In particular, if $ABC \neq 0$, then $(|A|, |B|, |C|)$ is an Euler brick.

Proof. Equation (3) follows by expanding (2). Using $v^2 = 5w^2 - u^2$, expansion of the left-hand side of (5) gives its right-hand side. The other two identities factor without further expansion:

$$\begin{aligned} A^2 + C^2 &= (v^2 - w^2)^2((u^2 - w^2)^2 + 4u^2w^2), \\ B^2 + C^2 &= 4u^2w^2(4v^2w^2 + (v^2 - w^2)^2). \end{aligned}$$

The bracketed factors are $(u^2 + w^2)^2$ and $(v^2 + w^2)^2$, respectively. Finally, by (5) and $v^2 - w^2 = 4w^2 - u^2$,

$$F = (u^4 - 5u^2w^2 - 4w^4)^2 + 4u^2w^2(u^2 - 4w^2)^2.$$

Expanding this last expression and collecting the factor $u^2 + 4w^2$ gives (8). $\square$

The binary form F is homogeneous of degree 16. Consequently, whether its reduction is zero, a nonzero square, or a nonsquare is well-defined on $\mathbb{P}^1(\mathbb{F}_p)$ for every odd prime p : rescaling (r, s) multiplies F by a sixteenth power.

3. A FIXED EXPONENT-ONE BLOCKER

Theorem 3.1 (Four blocked progressions). *Let $s = 1$ and*

$$r \equiv q \pmod{169}, \quad q \in \{3, 4, 7, 11\}, \quad r \geq 3.$$

Then the three expressions in (4) are nonzero and define an Euler brick. Its squared space diagonal satisfies

$$v_{13}(F(r, 1)) = 1.$$

It is therefore not a perfect cuboid. The same valuation remains true after division by the common divisor of the three edges.

Proof. Since $F(r, 1)$ is an integer polynomial in r , its value modulo 169 depends only on r modulo 169. Exact substitution gives Table 1. Every entry in the second column is divisible by 13 but not by 169, proving the valuation assertion. The last column records the three edge residues modulo 13; in particular, no edge and hence no common edge divisor is divisible by 13. Nonzero edge residues also exclude degeneracy. Lemma 2.1 now supplies the three integral face diagonals. $\square$

TABLE 1. The modulus-169 certificate and edge residues.

q	$F(q, 1) \bmod 169$	$F(q, 1)/13 \bmod 13$	$(A , B , C) \bmod 13$
3	52	4	(10, 9, 12)
4	39	3	(4, 1, 3)
7	52	4	(10, 4, 1)
11	117	9	(12, 10, 4)

Remark 3.2 (The first instance). At $(r, s) = (3, 1)$, the raw edge triple has common divisor 128. Dividing by it gives

$$(a, b, c) = (252, 275, 240), \quad (d_{ab}, d_{ac}, d_{bc}) = (373, 348, 365).$$

Thus

$$a^2 + b^2 + c^2 = 196729 = 13 \cdot 15133,$$

and the factor 13 occurs exactly once. Theorem 3.1 is uniform in the parameter rather than a check of isolated examples.

Remark 3.3. The phrase *exponent-one blocker* here means exactly a prime p with $v_p(A^2 + B^2 + C^2) = 1$. The universal Master-Hit conjecture in [4] additionally asks that a blocker be coprime to a specified list of canonical parameter expressions. Theorem 3.1 does not assert that stronger coprimality package.

4. A THREE-PRIME PROJECTIVE SIEVE

For an odd prime p , let QR_p be the set of nonzero quadratic residues in $\mathbb{F}_p$. Define

$$(9) \quad \mathcal{B}_p = \{[r : s] \in \mathbb{P}^1(\mathbb{F}_p) : F(r, s) \neq 0 \text{ and } F(r, s) \notin \text{QR}_p\}.$$

We label finite points by $t = r/s$ and the remaining point by $\infty = [1 : 0]$. The complete finite slope lists used below appear in Appendix A; in all three cases, $\infty \notin \mathcal{B}_p$.

Theorem 4.1 (Exact local sieve). *The bad-set cardinalities for the form (8) are*

$$|\mathcal{B}_{23}| = 16, \quad |\mathcal{B}_{31}| = 16, \quad |\mathcal{B}_{67}| = 40.$$

Let $M = 23 \cdot 31 \cdot 67$. Among the residue pairs $(r, s) \bmod M$ that are not simultaneously zero modulo any prime divisor of M , the proportion not excluded by these three tests is

$$\frac{24 - 16}{24} \cdot \frac{32 - 16}{32} \cdot \frac{68 - 40}{68} = \frac{7}{102}.$$

Hence the sieve proves that $F(r, s)$ is not a square on exactly 95/102 of the projective Chinese-remainder cells.

Proof. For each $p \in \{23, 31, 67\}$, evaluate the explicit form (8) at $(t, 1)$ for $t = 0, \dots, p-1$ and once at $(1, 0)$. Comparing the nonzero values with the set $\{x^2 : x \in \mathbb{F}_p^\times\}$ gives the lists in Appendix A, and therefore the stated cardinalities. If $F(r, s)$ were an integer square, then its reduction modulo every prime would be either zero or a quadratic residue. Every listed slope is consequently impossible.

A nonzero pair modulo p determines one of $p+1$ projective slopes, and every slope has exactly $p-1$ nonzero scalar representatives. The Chinese remainder theorem makes the three slope choices independent. Thus each triple of projective slopes has the same number $(22)(30)(66)$ of representing

pairs modulo M , and the displayed product is exact. Its complement is $1 - 7/102 = 95/102$. $\square$

Corollary 4.2. *Ordered primitive pairs $(r, s) \in \mathbb{Z}^2$, counted in expanding symmetric boxes, have limiting excluded proportion $95/102$ under the three-prime sieve.*

Proof. Every admissible residue pair modulo M has the same primitive-pair density. Indeed, Möbius inversion restricts the common divisor d to $(d, M) = 1$, and the resulting factor $\sum_{(d, M)=1} \mu(d)/d^2$ is independent of the chosen admissible residue class. The ratio is therefore the residue-class ratio of Theorem 4.1. $\square$

The theorem is a sieve, not a classification. A parameter cell surviving all three primes need not yield a square space norm; it has merely escaped these three local witnesses. Conversely, adding further primes can only shrink the surviving set, but the finite statement above is kept small enough to audit directly from the printed certificate.

5. EXACT CERTIFICATE AND REPRODUCIBILITY

The ancillary JSON certificate records the four modulus-169 evaluations, the full bad-slope sets, their cardinalities, and the example in Section 3. It is generated from the displayed formulas by the supplied certificate generator. The command

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python3 code/run_all_verifiers.py
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regenerates the data and runs two exact verifiers using only the Python standard library.

The primary verifier evaluates the factored form in (8). The independent verifier deliberately does not import that implementation: it reconstructs u, v, w and the three edge expressions, forms $A^2 + B^2 + C^2$ directly, and enumerates every projective point for each prime. It also checks signed parameters, the point at infinity, the face identities, positivity on sampled progression members, and preservation of the 13-adic obstruction under primitive scaling. All arithmetic used for the certificate is integer arithmetic; no numerical square-root, factorization heuristic, or elliptic-curve rank computation enters the proof.

Computer algebra and language-model assistance were used during exploratory algebra, literature triage, and manuscript preparation. The final statements were reduced to the displayed identities and finite residue tables, then checked by the two independent exact-arithmetic paths described above. The author reviewed the mathematical argument, scope, and citations.

6. SCOPE AND FURTHER QUESTIONS

The results establish a partial theorem for the parameter family (2)–(4). They do not prove that every Euler brick is represented by that family, so

they cannot imply global nonexistence of a perfect cuboid. The local sieve also leaves 7/102 of its chosen parameter cells unresolved.

The fixed-prime phenomenon suggests two concrete next steps. First, one can classify pairs (p, q) for which an entire class $r \equiv q \pmod{p^2}$ has $v_p(F(r, 1)) = 1$; this is a finite root-and-lift problem for each p . Second, the surviving projective cells can be carried into the genus-one or higher-genus models associated with this family, where local information may be combined with descent or a Mordell–Weil sieve. Such a combination would address structure in the survivors rather than merely increasing a height bound.

APPENDIX A. COMPLETE LOCAL RESIDUE LISTS

The entries below are the finite slopes $t \in \mathbb{F}_p$ for which $F(t, 1)$ is nonzero and nonsquare. The point at infinity is not bad for any of the three primes.

$$\begin{aligned}\mathcal{B}_{23} &= \{3, 4, 5, 6, 7, 9, 10, 12, 13, 14, 15, 16, 17, 18, 19, 21\}, \\ \mathcal{B}_{31} &= \{3, 4, 5, 6, 7, 10, 11, 13, 14, 17, 19, 20, 22, 23, 25, 26\}, \\ \mathcal{B}_{67} &= \{4, 6, 8, 9, 10, 11, 14, 15, 20, 21, 23, 24, 25, 27, 29, 30, \\ &\quad 31, 32, 34, 35, 37, 38, 41, 42, 43, 44, 48, 49, 50, 51, 52, 53, 54, \\ &\quad 56, 58, 59, 60, 61, 62, 65\}.\end{aligned}$$

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